AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 14 · WORKBOOK

Build a Human Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected classifier-construction record.

State Tower1's contract, build its paths, trace cases, construct Ribbon1, then inspect its validation claim.

Keep one habit: Keep input, contract result, model version and prediction or reason together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Learning from Examples

Repeating Efficiently

Features: Useful Clues

W1 · State the complete Tower1 contract

> Use the supplied reference and give accepted and rejected examples.

Your evidence record

- Tower1 reference: accept one whole-number height from 1 through 10 inclusive with unit cm. If any input check fails, record the original input, Tower1, NO PREDICTION and the reason, then stop that record. If accepted, predict tall at 6 cm or more and short otherwise, recording the original input, Tower1 and prediction. An accepted input is not proof that its prediction agrees with a comparison label.

- Write the complete Tower1 **input contract**. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted **input** does not have to be one of the examples used to fit the threshold.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Construct both outcome paths

> Write a procedure another learner can follow without inventing rules.

Your evidence record

- Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Trace accepted, boundary and rejected inputs

> Stop after rejection instead of coercing a blank value.

Your evidence record

- Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records **no prediction** for a blank, then continues and treats the blank as zero to report short.

  Identify the missing instruction and the proper final outcome.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Build an independent ribbon classifier

> Keep one first attempt before hints or feedback.

Read and explain independently

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build an independent ribbon classifier

MAIN LESSON · SECTION 16

W5 · Evaluate after the first Ribbon1 attempt

> Use the Ribbon1 contract from W4 and explain what validation leaves unknown.

Your evidence record

- Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown.

  Explain why changing a blank to 1 cm just to produce a category would be a mistake.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Make the sorter's current **decision boundary** visible as a rule chart or feature map.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Construct both outcome paths for Constructing a **classifier**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

Apply the rule and identify a limitation.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Use a human **classifier**: three straight sides forming a closed shape means triangle; four equal straight sides and four right angles forming a closed shape means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 14 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

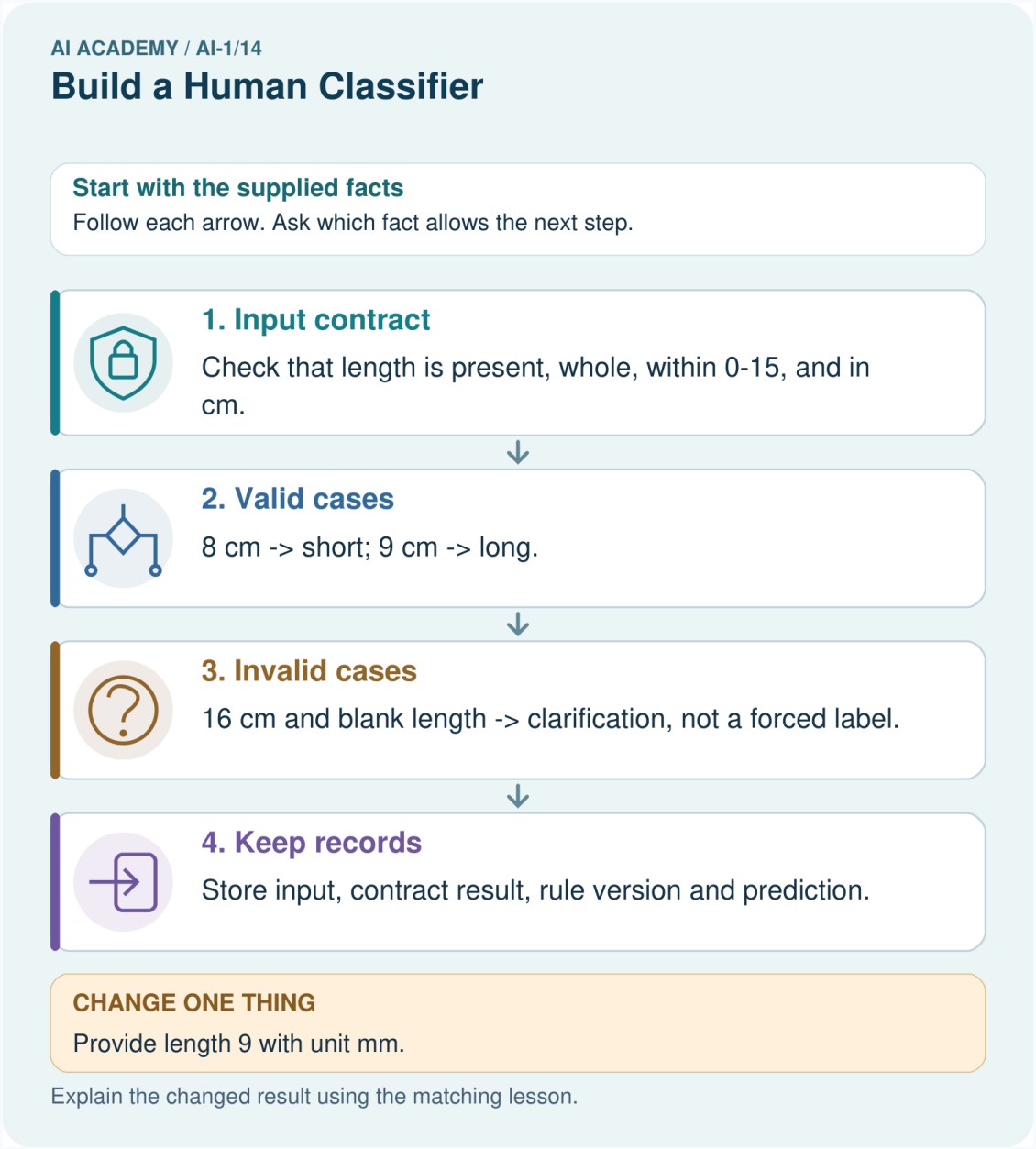
> Keep the explanation and the support you used.

Use this when helpful

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Valid **input**: whole-number length from 0 to 15 with unit cm.

Predict long at least 9 cm; otherwise short. Missing, wrong-unit or out-of-range values request clarification.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace 0 cm and explain why clarification is different from predicting short.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional ribbon sorter uses a fixed instruction card called RibbonR.
- Which labelled step matches this detail? Store **input**, contract result, rule version and **prediction**.
- 1. Invalid cases
- 2. **Input contract**

- 3. Keep records
- 4. Valid cases
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Keep records Store **input**, contract result, rule version and **prediction**. A **prediction** boundary answers a different question from **input validation**.

Length 9 passes the contract and meets the long boundary. Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Changed-case question

- Changed-case practice: Provide length 9 with unit mm. Which conclusion is supported by the supplied facts?
- 1. Request clarification rather than predicting long.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Request clarification rather than predicting long. RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Repair and save your work

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Build a Human Classifier: the case

AI ACADEMY / AI-1/14 · 01 · LOOK AT THE FACTS

Build a Human Classifier: the case

The situation

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Supplied evidence

Valid input: whole-number length from 0 to 15 with unit cm.
Predict long at least 9 cm; otherwise short.
Missing, wrong-unit or out-of-range values request clarification.

By the end, I can build a complete classifier that validates inputs before predicting.

Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional ribbon sorter uses a fixed instruction card called RibbonR. Supplied facts: Valid **input**: whole-number length from 0 to 15 with unit cm. Predict long at least 9 cm; otherwise short.

Missing, wrong-unit or out-of-range values request clarification.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



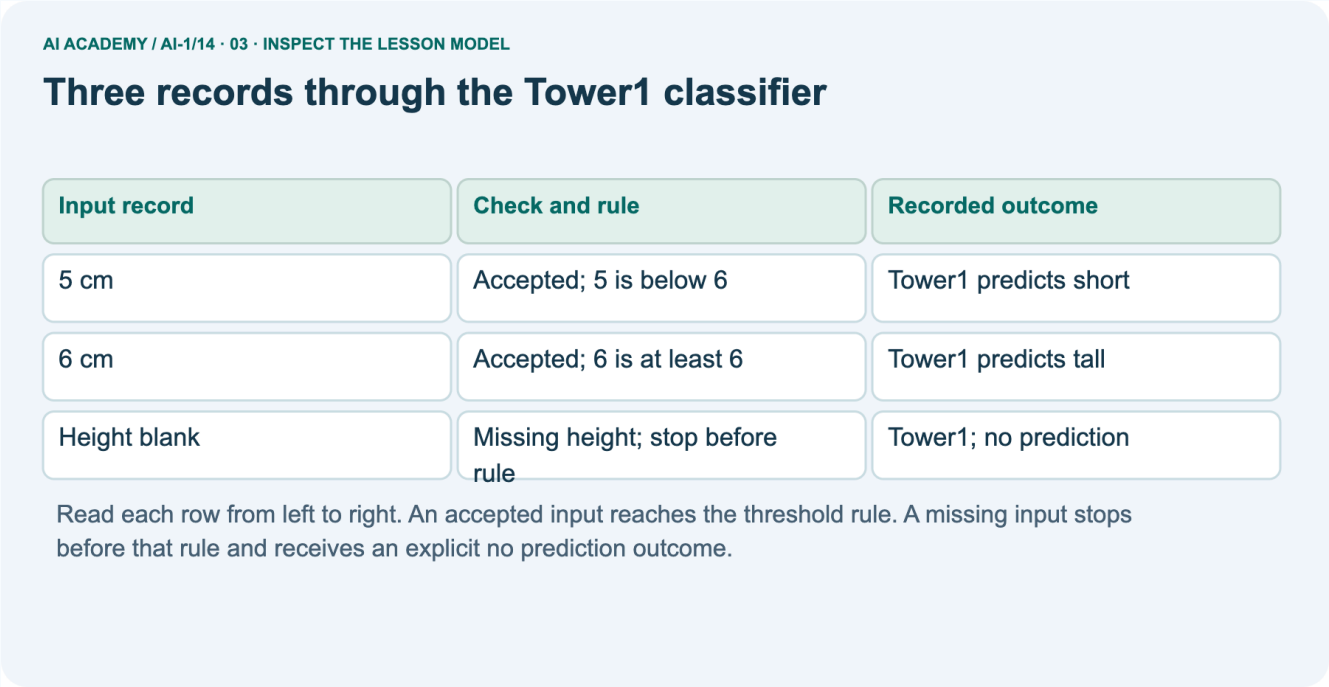
Follow the labelled picture with the matching explanation. Open the vector diagram

Input contract: Check that length is present, whole, within 0–15, and in cm. → Valid cases: 8 cm → short; 9 cm → long. → Invalid cases: 16 cm and blank length → clarification, not a forced label. → Keep records: Store **input**, contract result, rule version and **prediction**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three records through the Tower1 classifier



AI ACADEMY / AI-1/14 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

Follow the labelled picture with the matching explanation. Open the vector diagram

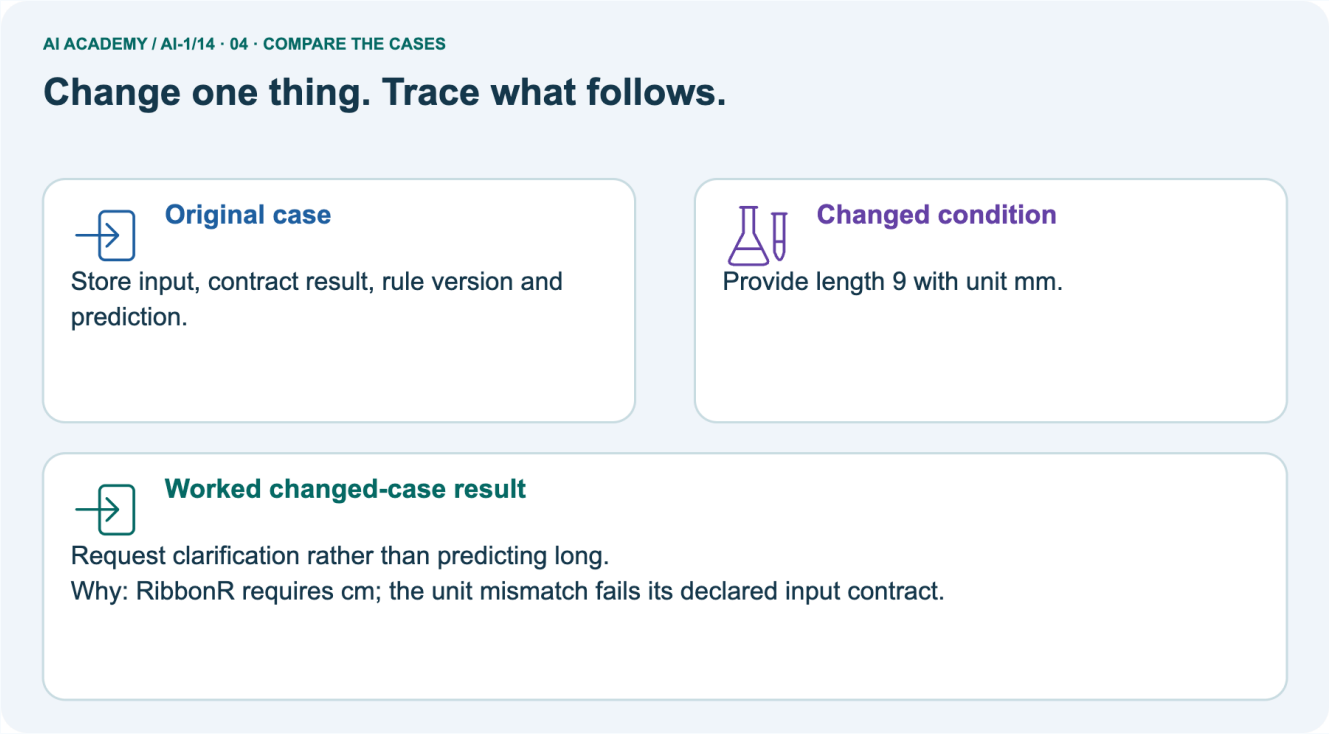
Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

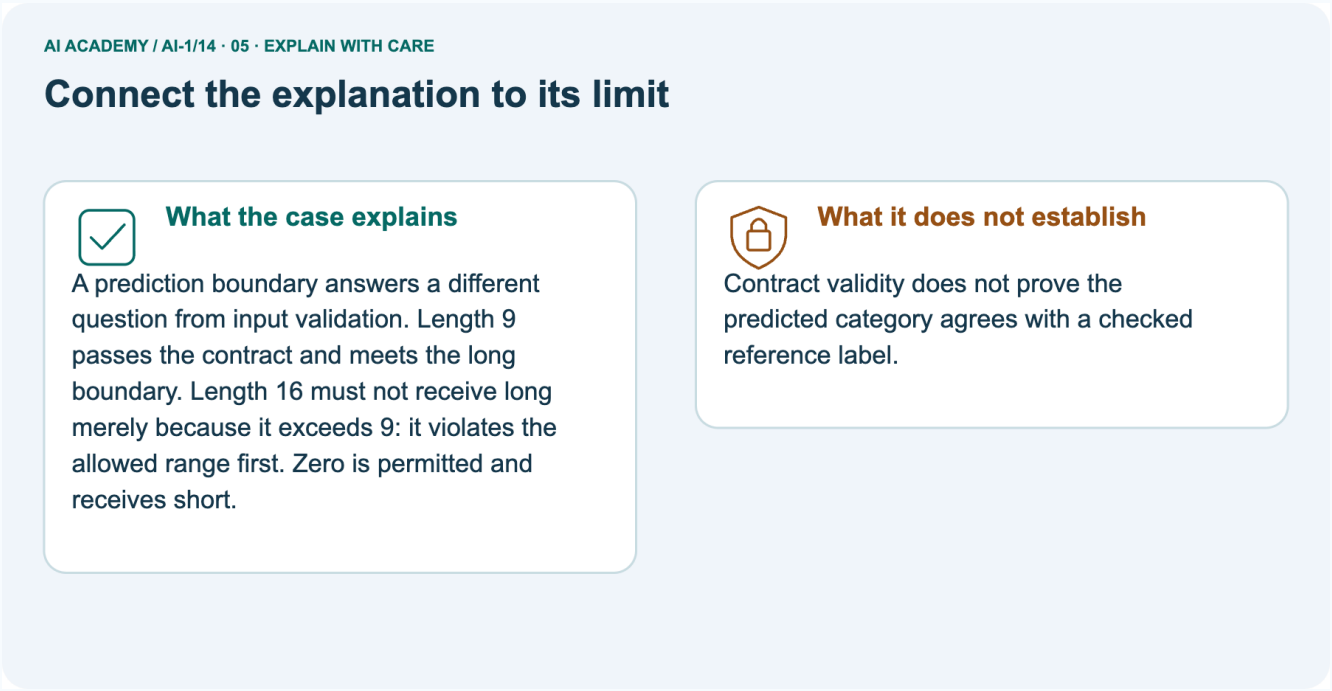
Original case: Store **input**, contract result, rule version and **prediction**. Changed condition: Provide length 9 with unit mm. Expected result: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

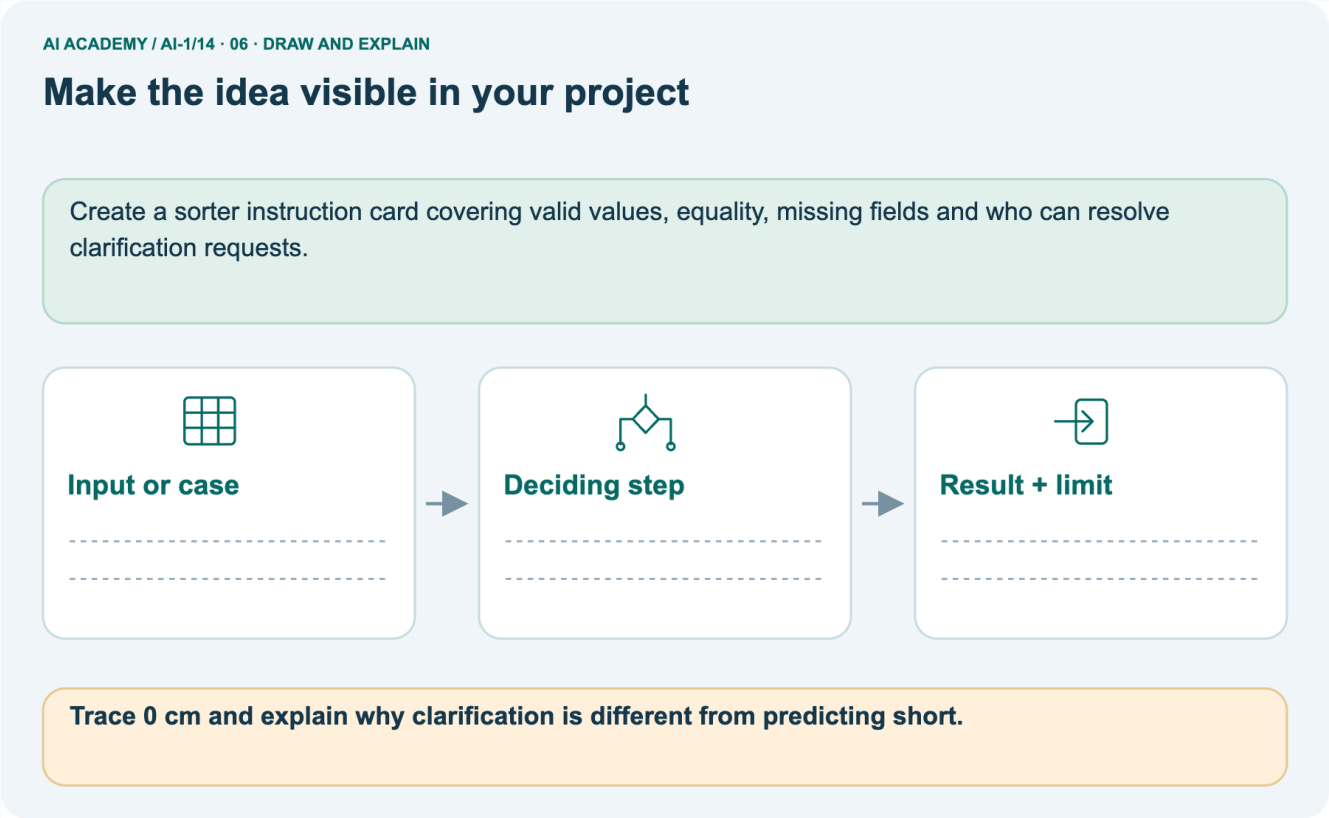
Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Limit: Contract validity does not prove the predicted category agrees with a checked reference label.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests. Draw your **input**, deciding step, result and limit.

Supported question: Trace 0 cm and explain why clarification is different from predicting short.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Build a Human Classifier: the case

02. Follow the deciding steps

03. Three records through the Tower1 classifier

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.